Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Assertion
$\cos^7 x + \sin^4 x = 1$ has only two solutions in the interval $-\pi, \pi$
Reason
$\cos^5 x + \cos^2 x - 2 = 0$ is possible only when $\cos x = 1$

(A) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion

(B) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion

(C) Assertion is correct but Reason is incorrect

(D) Assertion is incorrect but Reason is correct

Q.2

If $\tan\theta + \tan 4\theta + \tan 7\theta = \tan\theta \tan 4\theta \tan 7\theta$

(A) $\frac{n\pi}{4}, n \in Z$

(B) $\frac{n\pi}{7}, n \in Z$

(C) $\frac{n\pi}{12}, n \in Z$

(D) $n\pi, n \in Z$

Q.3

If α, β are different values of x satisfying $a\cos x + b\sin x = c$, then $\tan\left(\frac{\alpha+\beta}{2}\right) =$

(A) $a + b$

(B) $a - b$

(C) b/a

(D) a/b

Q.4

The values of θ lying between $\theta = 0$ and $\theta = \frac{\pi}{2}$ and satisfying the equation
$\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$, is

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(A) $\frac{11\pi}{24}, \frac{7\pi}{24}$

(B) $\frac{7\pi}{24}, \frac{5\pi}{24}$

(C) $\frac{5\pi}{24}, \frac{\pi}{24}$

(D) $\frac{\pi}{24}, \frac{11\pi}{24}$

Q.5

If $\sin 5x + \sin 3x + \sin x = 0$, then the value of x other than zero, lying between $0 \leq x \leq \frac{\pi}{2}$ is

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{12}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{4}$

Q.6

The most general value of θ which satisfies both the equations $\tan\theta = -1$ and $\cos\theta = 1/\sqrt{2}$ will be

(A) $n\pi + \frac{7\pi}{4}$

(B) $n\pi + (-1)^n \frac{7\pi}{4}$

(C) $2n\pi + \frac{7\pi}{4}$

(D) none of these

Q.7

If $1 + \sin\theta + \sin^2\theta + \ldots$ to $\infty = 4 + 2\sqrt{3}, 0 < \theta < \pi, \theta \neq \frac{\pi}{2}$, then $\theta =$

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{3}$ or, $\frac{\pi}{6}$

(D) $\frac{\pi}{3}$ or, $\frac{2\pi}{3}$

Q.8

The general solution of the equation $\sin 2x + 2\sin x + 2\cos x + 1 = 0$, is

(A) $3n\pi - \frac{\pi}{4}, n \in Z$

(B) $2n\pi + \frac{\pi}{4}, n \in Z$

(C) $2n\pi + (-1)^n \sin^{-1}\left(\frac{1}{\sqrt{3}}\right), n \in Z$

(D) $n\pi - \frac{\pi}{4}, n \in Z$

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Q.9

The solution of the equation $\cos^2\theta + \sin\theta + 1 = 0$, lies in the interval

(A) $(-\pi/4, \pi/4)$

(B) $(\pi/4, 3\pi/4)$

(C) $(3\pi/4, 5\pi/4)$

(D) $(5\pi/4, 7\pi/4)$

Q.10

The least positive non-integral solution of the equation
$\sin\pi\left(x^2 + x\right) = \sin\pi x^2$ is

(A) rational

(B) irrational of the form $\sqrt{\mathrm{p}}$

(C) irrational of the form $\frac{\sqrt{\mathrm{p}}-1}{4}$, where p is an odd integer

(D) irrational of the form $\frac{\sqrt{\mathrm{p}}+1}{4}$, where pis an even integer

Q.11

If $\cos\theta + \cos 2\theta + \cos 3\theta = 0$, then the general
value of θ is :

(A) $\theta = 2m\pi \pm 2\pi/3$

(B) $\theta = 2m\pi \pm \pi/4$

(C) $\theta = m\pi + (-1)^n 2\pi/3$

(D) $\theta = m\pi + (-1)^n \pi/3$

Q.12

Which one of the following is one of the solutions of the equation of the equation $\tan 2\theta \cdot \tan\theta = 1$?

(A) $\pi/12$

(B) $\pi/6$

(C) $\pi/4$

(D) $\pi/3$

Q.13

General solution of the equation $(\sqrt{3}-1)\sin\theta + (\sqrt{3}+1)\cos\theta = 2$ is

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(A) $2n\pi \pm \frac{\pi}{4} + \frac{\pi}{12}$

(B) $n\pi + (-1)^n \frac{\pi}{2}$

(C) $2n\pi \pm \frac{\pi}{4} - \frac{\pi}{12}$

(D) None

Q.14

The solution set of the system of equation $x + y = 2\pi/3, \cos x + \cos y = 3/2$, where x and y are real, is

(A) $x = \frac{\pi}{3} - n\pi, y = n\pi$

(B) ϕ

(C) $x = n\pi, y = \frac{\pi}{3} - n\pi$

(D) None of these

Q.15

Let n be a fixed positive integer such that $\sin\left(\frac{\pi}{2n}\right) + \cos\left(\frac{\pi}{2n}\right) = \frac{\sqrt{n}}{2}$, then :

(A) $\mathrm{n} = 4$

(B) $n = 5$

(C) $\mathrm{n} = 6$

(D) None of these

Answers & Solutions

Q.1 Answer:

Assertion is incorrect but Reason is correct

Solution:

statement 1.

$\cos^7 x + \sin^4 x = 1$

$\Rightarrow \cos^7 x = 1 - \sin^4 x = \left(1 - \sin^2 x\right)\left(1 + \sin^2 x\right) = \cos^2 x\left(2 - \cos^2 x\right)$

$\Rightarrow \cos^2 x\left(\cos^5 x + \cos^2 x - 2\right) = 0$

$\cos x = 0 \Rightarrow x = -\frac{\pi}{2}, \frac{\pi}{2}$

or, $\cos^5 x + \cos^2 x = 2$ we know $-1 \leq \cos x \leq 1$

thus $\cos x = 1$ is the only solution $\Rightarrow x = 0$

Hence number of solution is 3 .

Therefore statement 1 is incorrect.

statement 2.

$\cos^5 x + \cos^2 x = 2$

we know $-1 \leq \cos x \leq 1$

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Hence this equation is only solvable if $\cos x = 1$

therefore statement 2 is correct.

Q.2 Answer:

$\frac{n\pi}{12}, n \in Z$

Solution:

We have, $\tan\theta + \tan 4\theta + \tan 7\theta = \tan\theta \tan 4\theta \tan 7\theta$

$\tan\theta + \tan 4\theta = -\tan 7\theta(1 - \tan\theta \tan 4\theta)$

$\frac{\tan\theta + \tan 4\theta}{1 - \tan\theta \tan 4\theta} = -\tan 7\theta$

$\tan(\theta + 4\theta) = -\tan 7\theta -$

Since, $\frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan(A + B)$

$\tan 5\theta = \tan(-7\theta)$

$\therefore 5\theta = (-7\theta)$

$12\theta = 0$ **or** $12\theta = n\pi$

$\therefore \theta = \frac{n\pi}{12}, \quad \forall n$ **as I**

Q.3 Answer:

b/a

Solution:

clearly a $a\cos\alpha + b\sin\alpha = c = a\cos\beta + b\sin\beta$

$a(\cos\alpha - \cos\beta) + b(\sin\alpha - \sin\beta) = 0$

$a\left(2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}\right) = b\left(2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}\right)$

$\tan\frac{\alpha+\beta}{2} = \frac{b}{a}$

Q.4 Answer:

$\frac{11\pi}{24}, \frac{7\pi}{24}$

Solution:

$$\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix}$$

$C_1 \to C_1 + C_2$

$$\begin{vmatrix} 1+\sin^2\theta+\cos^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta+\cos^2\theta+1 & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta+\cos^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1+1 & \cos^2\theta & 4\sin 4\theta \\ 1+1 & 1+\cos^2\theta & 4\sin 4\theta \\ 1 & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 2 & \cos^2\theta & 4\sin 4\theta \\ 2 & 1+\cos^2\theta & 4\sin 4\theta \\ 1 & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix}$$

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$R_2 : R_2 \to R_2 - R_1 \quad R_3 \to R_3 - R_1$

$$\Rightarrow \begin{vmatrix} 2 & \cos^2\theta & 4\sin 4\theta \\ 2-2 & 1+\cos^2\theta-\cos^2\theta & 4\sin 4\theta - 4\sin 4\theta \\ 2-2 & \cos^2\theta - \cos^2\theta & 1+4\sin 4\theta - 4\sin 4\theta \end{vmatrix} = 0$$

$$\begin{vmatrix} 2 & \cos^2\theta & 4\sin 4\theta \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0$$

$\Rightarrow 0 + (1)[2 - (-1)(4\sin 4\theta)] = 0$

$\Rightarrow 2 + 4\sin 4\theta = 0$

$\Rightarrow \sin 4\theta = \frac{-2}{4}$

$\Rightarrow \sin 4\theta = \frac{-1}{2}$

$\Rightarrow 4\theta = \pi + \frac{\pi}{6}, 4\theta = 2\pi - \frac{\pi}{6}$ [$\because$ **only in 3rd and 4th quadrants sin is negative**]

$\Rightarrow 4\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$

$\theta = \frac{7\pi}{24}, \frac{11\pi}{24}$

Q.5 Answer:

$\frac{\pi}{3}$

Solution:

$\sin 5x + \sin 3x + \sin x = 0$

$(\sin 5x + \sin x) + \sin 3x = 0$

$\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$

$2\sin 3x \cdot \cos 2x + \sin 3x = 0$

$\sin 3x(2\cos 2x + 1) = 0$

$\sin 3x = \sin 0 \quad$ or $\quad 2\cos 2x + 1 = 0$

$3x = n\pi \quad \cos 2x = \cos 2\pi/3$

$x = \frac{n\pi}{3} \quad 2x = 2\pi/3 + 2n\pi$

$0 \le x \le \pi/2 \quad x = \pi/3 \pm n\pi.$

$x = n\pi/3$ gives, $x = 0, \pi/3$

Q.6 Answer:

$2n\pi + \frac{7\pi}{4}$

Solution:

$\tan\theta = -1$ **and** $\cos\theta = \frac{1}{\sqrt{2}}$

Common solutions lie in 4th quadrant which is $\theta = \frac{7\pi}{4}$

Hence general solution is $\theta = 2n\pi + \frac{7\pi}{4}$, wheren $\in$ I

Q.7 Answer:

$\frac{\pi}{3}$ or, $\frac{2\pi}{3}$

Solution:

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$1+\sin x+\sin^2 x+\sin^3 x+\ldots\infty=4+2\sqrt{3}$
$\Rightarrow \frac{1}{1-\sin x}=4+2\sqrt{3}$
$\Rightarrow 1-\sin x=\frac{1}{4+2\sqrt{3}} * \frac{4-2\sqrt{3}}{4-2\sqrt{3}}$
$\Rightarrow 1-\sin x=\frac{4-2\sqrt{3}}{16-12}=\frac{4-2\sqrt{3}}{4}$
$\Rightarrow 1-\sin x=1-\frac{\sqrt{3}}{2}$
$\Rightarrow \sin x=\frac{\sqrt{3}}{2}$
$\therefore x=\frac{\pi}{3}$ **or** $\frac{2\pi}{3}$.

Q.8 Answer:

$n\pi-\frac{\pi}{4}, n\in Z$

Solution:

We have, $\sin 2x+2\sin x+2\cos x+1=0$
$\Rightarrow \quad (1+2\sin x\cos x)+2(\sin x+\cos x)=0$
$\Rightarrow \quad (\sin x+\cos x)^2+2(\sin x+\cos x)=0$
$\Rightarrow \quad (\sin x+\cos x)(\sin x+\cos x+2\ |=0$
$\Rightarrow \quad \sin x+\cos x=0 \quad \left[\begin{array}{l} \because -\sqrt{2}\leq \sin x+\cos x\leq 1 \\ \therefore \sin x+\cos x\neq -2 \end{array}\right]$
$\Rightarrow \quad \tan x=-1$
$\Rightarrow \quad x=n\pi-\frac{\pi}{4}, n\in Z$

Q.9 Answer:

$(5\pi/4, 7\pi/4)$

Solution:

We have, $\cos^2\theta+\sin\theta+1=0\Longrightarrow 1-\sin^2\theta+\sin\theta+1=0$
$\Longrightarrow (1-\sin\theta)(1+\sin\theta)+(1+\sin\theta)=0$
$\Longrightarrow (1-\sin\theta+1)(1+\sin\theta)=0$
$\Longrightarrow \sin\theta=-1(\because \sin\theta\neq 2)$
$\Longrightarrow \theta=3\pi/2$
$\therefore \theta$ **lies in the interval** $\left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$

Q.10 Answer:

rational

Solution:

We have, $\sin\pi\left(x^2+x\right)=\sin\pi x^2$
$\Rightarrow \pi\left(x^2+x\right)=n\pi+(-1)^n\pi x^2$
$\therefore$ **Either** $x^2+x=2m+x^2\Rightarrow x=2m\in I$
or $x^2+x=k-x^2$**, where** k **is an odd integer** $\Rightarrow 2x^2+x-k=0\Rightarrow x=\frac{-1\pm\sqrt{1+8k}}{4}$

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For least positive non-integral solution
is $x = \frac{1}{2}$, **when** $k = 1$

Q.11 Answer:

$\theta = 2m\pi \pm 2\pi/3$

Solution:

Given $\cos\theta + \cos 2\theta + \cos 3\theta = 0$
$\Rightarrow (\cos 3\theta + \cos\theta) + \cos 2\theta = 0$
$\Rightarrow 2\cos 2\theta \cdot \cos\theta + \cos 2\theta = 0$
$\Rightarrow \cos 2\theta \cdot (2\cos\theta + 1) = 0$
we have, $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$
For general value of θ, $\cos 2\theta = 0$
$\Rightarrow \quad \cos 2\theta = \cos\frac{\pi}{2} \Rightarrow \quad 2\theta = 2m\pi \pm \frac{\pi}{2}$
$\Rightarrow \quad \theta = m\pi \pm \frac{\pi}{4}$ **or** $2\cos\theta + 1 = 0$
$\Rightarrow \quad \cos\theta = \frac{-1}{2} \Rightarrow \cos\theta = \cos\frac{2\pi}{3}$
So, $\theta = 2m\pi \pm \frac{2\pi}{3}$

Q.12 Answer:

$\pi/6$

Solution:

$\tan 2\theta \cdot \tan\theta = 1$
$\Rightarrow \frac{2\tan\theta}{1-\tan^2\theta} \cdot \tan\theta = 1$
$\Rightarrow 2\tan^2\theta = 1 - \tan^2\theta \Rightarrow 3\tan^2\theta = 1$
$\Rightarrow \tan^2\theta = \frac{1}{3} = \left(\frac{1}{\sqrt{3}}\right)^2$
$\Rightarrow \tan^2\theta = \tan^2(30^\circ) = \tan^2\left(\frac{\pi}{6}\right)$
$\left[\because \theta = n\pi \pm \frac{\pi}{6}\right]; \therefore \theta = \frac{\pi}{6}$

Q.13 Answer:

$2n\pi \pm \frac{\pi}{4} + \frac{\pi}{12}$

Solution:

Let $\sqrt{3}+1 = r\cos\alpha$, **and** $\sqrt{3}-1 = r\sin\alpha$
$r^2 = (\sqrt{3}+1)^2 + (\sqrt{3}-1)^2 = 8$ **i.e.** $\alpha = \pi/12$
From the equation, $r\cos(\theta - \alpha) = 2$
$\Rightarrow \quad \cos(\theta - \pi/12) = 1/\sqrt{2} = \cos(\pi/4)$
$\theta = 2n\pi \pm \pi/4 + \pi/12$

Q.14 Answer:

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Solution:

We have $\cos x + \cos y = \frac{3}{2}$
$\Rightarrow \quad 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) = \frac{3}{2}$
$\Rightarrow \quad \cos\left(\frac{x-y}{2}\right) = \frac{3}{2}\left(\because x+y = \frac{2\pi}{3}\right)$
Which is not possible $\left(\text{as } \cos\theta \leq 1\right)$ Thus, the solution set is a null set.

Q.15 Answer:

$n = 6$

Solution:

$\sin\frac{\pi}{2n} + \cos\frac{\pi}{2n} = \sqrt{2}\sin\left(\frac{\pi}{4} + \frac{\pi}{2n}\right)$
$\Rightarrow \frac{\sqrt{n}}{2} = \sqrt{2}\sin\left(\frac{\pi}{4} + \frac{\pi}{2n}\right)$
So, for $n > 1$, $\frac{\sqrt{n}}{2\sqrt{2}} = \sin\left(\frac{\pi}{4} + \frac{\pi}{2n}\right) > \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$
Thus, $n > 4$ Since, $\sin\left(\frac{\pi}{4} + \frac{\pi}{2n}\right) < 1$ for all $n > 2$, we get
$\frac{\sqrt{n}}{2\sqrt{2}} < 1$ or $n < 8$, So that $4 < n < 8$. By actual
verification we find that only $n = 6$ satisfies the given relation.

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